

DATA ENGINEER ROAD MAP

A comprehensive guide for transitioning into Data Engineering
from any background

Covering: Freshers | Data Analysts | Other Domains

Includes: Learning Paths | Projects | Architecture Diagrams

SECTION 1: FRESHERS (0 Experience) -> Data Engineer

Timeline: 4-6 Months (Intensive)

Phase	Duration	Topics	What to Learn (Topics)
Phase 1	Week 1-3	Programming Fundamentals	Python: variables, data types, loops, functions, file handling, list/dict comprehensions, error handling, reading CSV/JSON SQL: SELECT, WHERE, GROUP BY, JOINS, subqueries, window functions, CTEs, HAVING, ORDER BY, CASE statements Linux/CLI: cd, ls, grep, cat, pipe, chmod, cron, ssh basics Git: init, add, commit, push, pull, branch, merge, .gitignore
Phase 2	Week 3-7	Databases, Data Modeling & Warehousing	PostgreSQL: CREATE TABLE, constraints, indexes, EXPLAIN, transactions, views, stored procedures Data Modeling: normalization (1NF, 2NF, 3NF), ER diagrams, primary/foreign keys, one-to-many/many-to-many relationships Warehousing: star schema, snowflake schema, fact tables, dimension tables, SCD Type 1/2, OLTP vs OLAP, partitioning, columnar storage basics Tools: PostgreSQL locally, BigQuery or Snowflake (free tier)
Phase 3	Week 7-11	ETL/ELT & Orchestration	ETL concepts: extract, transform, load, batch vs streaming, full load vs incremental load, idempotency, backfills Apache Airflow: DAGs, tasks, operators, scheduling, XComs, task dependencies, retries, sensors, connections dbt: models, sources, refs, tests, seeds, snapshots, materializations (table/view/incremental), Jinja templating Data Quality: null checks, schema validation, row counts, freshness checks, Great Expectations basics
Phase 4	Week 11-16	Big Data & Cloud	PySpark: RDDs, DataFrames, transformations (map, filter, groupBy, join), actions, partitioning, caching, spark-submit Kafka basics: producers, consumers, topics, partitions, offsets, consumer groups, basic streaming concepts AWS: S3 (buckets, prefixes, lifecycle), IAM basics, Glue (crawlers, jobs), Redshift basics, Lambda triggers OR GCP: GCS, BigQuery, Dataflow, Cloud Functions
Phase 5	Week 16-19	DevOps Basics	Docker: Dockerfile, images, containers, docker-compose, volumes, port mapping, building data pipeline images CI/CD basics: GitHub Actions (yaml workflows), running tests on push, linting, auto-deploy concepts Terraform basics: providers, resources, plan/apply, provisioning S3 buckets or cloud resources via code
Phase 6	Week 19-24	Projects & Interview Prep	Easy projects to build: 1. Weather API -> Airflow DAG -> PostgreSQL -> daily refresh 2. CSV sales data -> dbt transforms -> star schema warehouse 3. Web scraper -> S3 raw files -> Spark clean -> Redshift 4. GitHub API -> incremental load -> analytics dashboard Interview prep: SQL problems (LeetCode/HackerRank), system design basics, Python coding rounds, resume with GitHub links

SECTION 2: DATA ANALYST -> Data Engineer

Timeline: 2-3 Months (Intensive, leveraging existing SQL, data, and domain knowledge)

Advantages You Already Have:

- SQL proficiency
- Business domain knowledge
- Understanding of data quality issues
- Stakeholder communication

Phase	Duration	Topics	What to Learn (Topics)
Phase 1	Week 1-3	Advanced Python & Data Modeling	Python: OOP (classes, inheritance), decorators, generators, context managers, unit testing (pytest), virtual environments Data Modeling: dimensional modeling, star/snowflake schema, fact vs dimension tables, SCD Type 1/2, partitioning, indexing strategies, OLTP vs OLAP, denormalization Tools: PostgreSQL advanced (window functions, CTEs, EXPLAIN), BigQuery or Snowflake free tier setup
Phase 2	Week 3-6	ETL/ELT Pipelines	Apache Airflow: DAGs, operators, scheduling, XComs, sensors, connections, retries, task dependencies, backfills dbt: models, sources, refs, tests, seeds, snapshots, incremental materializations, Jinja templating Concepts: full load vs incremental, idempotency, upserts, data quality checks (row counts, null checks, freshness)
Phase 3	Week 6-9	Cloud & Big Data	PySpark: DataFrames, transformations (filter, groupBy, join), actions, partitioning, caching, reading from S3/GCS AWS: S3, Glue (crawlers, ETL jobs), Redshift, Lambda OR GCP: GCS, BigQuery, Dataflow, Cloud Functions Kafka basics: producers, consumers, topics, consumer groups
Phase 4	Week 9-11	DevOps Basics	Docker: Dockerfile, images, containers, docker-compose, volumes, containerizing Airflow + dbt pipelines CI/CD basics: GitHub Actions workflows, running tests on push, linting SQL/Python, auto-deploy concepts
Phase 5	Week 11-12	Projects & Interview Prep	Easy projects to build: 1. Automate an existing Excel report -> Airflow + dbt + DW 2. API ingestion -> incremental load -> analytics tables 3. CSV pipeline -> data quality checks -> star schema Interview prep: explain pipeline design decisions, system design (design a reporting pipeline), SQL optimization, discuss trade-offs (batch vs streaming, star vs snowflake)

SECTION 3: OTHER DOMAINS -> Data Engineer

Timeline: 3-4 Months (Intensive, varies by background)

Phase	Duration	Topics	What to Learn (Topics)
Phase 1	Week 1-4	SQL & Python for Data	SQL: SELECT, JOINS (inner, left, full), GROUP BY, window functions (ROW_NUMBER, RANK, LAG/LEAD), CTEs, subqueries, CASE statements, aggregate functions, HAVING Python: variables, loops, functions, file I/O, pandas basics (read_csv, filtering, groupby, merge), requests library, JSON parsing, error handling, virtual environments Data formats: CSV, JSON, Parquet, Avro (when to use which)
Phase 2	Week 4-7	Databases, Data Modeling & Warehousing	PostgreSQL: CREATE TABLE, constraints, indexes, views, EXPLAIN Data Modeling: normalization (1NF, 2NF, 3NF), ER diagrams, star schema, snowflake schema, fact vs dimension tables, SCD Type 1/2, OLTP vs OLAP, partitioning basics Warehousing: columnar storage, BigQuery or Snowflake free tier, loading data into warehouse, basic query optimization
Phase 3	Week 7-10	ETL/ELT & Orchestration	ETL concepts: extract, transform, load, full vs incremental, idempotency, upserts, backfills, batch vs streaming Apache Airflow: DAGs, tasks, operators, scheduling, XComs, connections, retries, sensors, task dependencies dbt: models, sources, refs, tests, materializations (table/view/incremental), seeds, snapshots
Phase 4	Week 10-13	Big Data & Cloud	PySpark: DataFrames, transformations (map, filter, groupBy, join), actions, partitioning, caching, spark-submit Kafka basics: producers, consumers, topics, partitions, consumer groups, basic streaming concepts AWS: S3, IAM basics, Glue, Redshift, Lambda OR GCP: GCS, BigQuery, Dataflow, Cloud Functions
Phase 5	Week 13-15	DevOps Basics	Docker: Dockerfile, images, containers, docker-compose, volumes, containerizing data pipelines CI/CD basics: GitHub Actions workflows, running tests on push, linting, auto-deploy concepts Git workflow: branching strategies, pull requests, code review
Phase 6	Week 15-16	Projects & Interview Prep	Easy projects to build: 1. Public API -> Airflow DAG -> PostgreSQL -> daily refresh 2. CSV files -> dbt transforms -> star schema in BigQuery 3. Web scraper -> S3 -> PySpark clean -> warehouse Interview prep: SQL problems (LeetCode/HackerRank), explain pipeline architecture, Python coding, system design basics, resume with GitHub project links

Background-Specific Advantages:

Previous Role	Transferable Skills
Software Engineer	Programming, system design, testing, CI/CD
DevOps/SRE	Cloud, Docker, IaC, monitoring, automation
QA Engineer	Testing frameworks, data validation, attention to detail
Non-tech (Finance, Marketing)	Domain expertise, business logic, stakeholder needs

COMMON RESOURCES ACROSS ALL PATHS

Category	Recommended Tools/Platforms
Practice SQL	StrataScratch, DataLemur, LeetCode, HackerRank, Ankit Bansal (YouTube), SQLZoo
Learn Cloud	AWS/GCP free tier, cloud certifications
Build Projects	Kaggle datasets, public APIs, personal data
Communities	r/dataengineering, Data Engineering Weekly, dbt Community
Certifications	AWS Data Engineer Associate, GCP Professional Data Engineer, Databricks

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SAMPLE DATA ENGINEERING PROJECTS

Beginner Projects (Good for Freshers & Domain Switchers)

#	Project Name	Description	Tech Stack
1	Web Scraping to Database Pipeline	Scrape job listings/weather data from websites, clean and store in a database. Schedule daily refreshes.	Python, BeautifulSoup, PostgreSQL, cron jobs
2	CSV to Data Warehouse Loader	Ingest multiple CSV files (e.g., sales data), transform, apply star schema, and load into a warehouse.	Python, pandas, SQL, dbt, BigQuery/PostgreSQL
3	API Data Ingestion Pipeline	Pull data from a public API (Spotify, Twitter, OpenWeather), transform it, and store incrementally.	Python, requests, Airflow, PostgreSQL/S3
4	File Format Converter	Build a tool that converts between CSV, JSON, Parquet, and Avro with schema validation.	Python, PyArrow, PySpark
5	Database Change Data Capture	Track changes in a source database and replicate inserts/updates/deletes to a target database.	Python, PostgreSQL, Debezium, Kafka

Intermediate Projects (Good for Data Analysts Transitioning)

#	Project Name	Description	Tech Stack
6	Automated Reporting Pipeline	Replace manual Excel reports with an automated pipeline: ingest -> transform -> quality checks -> dashboard.	Airflow, dbt, BigQuery/Redshift, Great Expectations
7	E-Commerce Data Warehouse	Build a complete warehouse with fact tables (orders, clicks) and dimension tables (products, customers).	dbt, Snowflake/BigQuery, Airflow, Looker/Metabase
8	Real-Time Stock Dashboard	Stream stock price data, compute moving averages & alerts, and serve to a live dashboard.	Kafka, PySpark Streaming, InfluxDB, Grafana
9	Data Lake (Bronze/Silver/Gold)	Build a 3-layer data lake with schema evolution and partitioning strategies.	AWS S3/GCS, Spark, Delta Lake/Hudi, Airflow
10	SCD Type 2 Implementation	Implement slowly changing dimensions for customer/product data to track historical changes.	dbt, Snowflake/Redshift, SQL, Airflow

Advanced Projects (Portfolio-Worthy / Interview-Ready)

#	Project Name	Description	Tech Stack
11	End-to-End Streaming Pipeline	Ingest clickstream events, process in real-time, aggregate, and serve for analytics.	Kafka, Spark Streaming, Cassandra/Druid, Docker, K8s
12	Multi-Source Data Integration	Combine data from APIs, databases, flat files, and streams into a unified warehouse with lineage.	Airflow, Spark, dbt, Snowflake, Docker
13	IoT Sensor Data Pipeline	Process millions of sensor readings/min, detect anomalies, store time-series, trigger alerts.	Kafka, Spark, AWS IoT/Kinesis, TimescaleDB, Grafana
14	Pipeline with Full CI/CD	Production-grade pipeline with tests, automated deployments, monitoring, and alerting.	Airflow, dbt, Docker, GitHub Actions, Terraform
15	Event-Driven Data Mesh	Decentralized data platform where domains own their data products with governance.	Kafka, Spark, K8s, dbt, Snowflake, Datahub

Project Ideas by Open-Source APIs (Quick Reference)

Open-Source API	Project Idea
OpenWeatherMap API (free tier)	Weather data warehouse with forecasting and historical trend analysis
GitHub REST/GraphQL API	Developer activity tracker & repo analytics
Open-Meteo API (free, no key)	Global climate data pipeline with geospatial processing and alerting
CoinGecko API (free, no key)	Cryptocurrency price tracking, aggregation, and historical trend warehouse
JSONPlaceholder API (free)	Practice ETL pipeline with mock user/post/comment data (beginners)
REST Countries API (free)	Country demographics data warehouse with regional analytics
OpenLibrary API (free)	Book metadata pipeline with author and publishing trend analytics
PokeAPI (free, no key)	Practice data modeling with nested/complex JSON transformations
NASA Open APIs (free key)	Space event data ingestion (asteroids, Mars rover photos, solar flares)
World Bank Open Data API	Global economic indicators pipeline with multi-country comparisons
OpenFoodFacts API (free)	Nutrition data lake with product categorization and health score analytics
USGS Earthquake API (free)	Real-time earthquake event streaming pipeline with geospatial visualization
NewsAPI.org (free tier)	News article ingestion with topic classification and trend detection
OpenAQ API (free)	Air quality monitoring pipeline across global cities with alerting
RandomUser API (free)	Synthetic user data pipeline for testing SCD Type 2 and quality frameworks

DATA FLOW ARCHITECTURES

Project 1: Web Scraping to Database Pipeline

```

[Website/HTML Pages]
  |
  | (BeautifulSoup / Scrapy)
  v
[Raw HTML/JSON] --> [Python Script: Parse & Extract]
  |
  | (Clean, deduplicate, validate)
  v
[Cleaned Data (DataFrames)]
  |
  | (psycopg2 / SQLAlchemy)
  v
[PostgreSQL Database]
  |
  | (cron job - daily schedule)
  v
[Incremental Updates]

```

Project 2: CSV to Data Warehouse Loader

```

[Multiple CSV Files (Sales, Products, Customers)]
  |
  | (Python / pandas read_csv)
  v
[Raw DataFrames]
  |
  | (Data cleaning: nulls, types, dedup)
  v
[Staging Tables (raw schema)]
  |
  | (dbt transformations)
  v
[Star Schema]
+-- [Fact Table: fact_sales]
+-- [Dim Table: dim_products]
+-- [Dim Table: dim_customers]
+-- [Dim Table: dim_date]
  v
[BigQuery / PostgreSQL Warehouse] --> [Dashboard]

```

Project 3: API Data Ingestion Pipeline

```

[Public API (OpenWeatherMap / GitHub)]
  |
  | (HTTP GET - Python requests)
  v
[JSON Response]
  |
  | (Parse, flatten nested JSON)
  v
[Raw Layer (S3 / local storage)]
  |
  | (Airflow DAG - scheduled extraction)
  v
[Transform: clean, enrich, aggregate]
  |
  | (Incremental load - upsert logic)
  v
[PostgreSQL / Data Warehouse]

```

Project 6: Automated Reporting Pipeline

```

[Data Sources: DB, APIs, Files]
  |
  | (Airflow DAG - Extract task)
  v
[Raw/Staging Layer]
  |
  | (dbt models - Transform task)
  v
[Cleaned & Joined Tables]
  |
  | (Great Expectations - Quality checks)
  v
[Validated Data] --- (fail) --> [Alert: Slack/Email]
  |
  | (pass)
  v
[Aggregated Reporting Tables]
  |
  v
[Dashboard (Metabase / Looker / Tableau)]

```

Project 9: Data Lake (Bronze/Silver/Gold)

```

[Source Systems: APIs, DBs, Files, Streams]
  |
  v
+=====+
| BRONZE LAYER (Raw)          |
| - Raw as-is, partitioned   |
| - Append-only, immutable    |
+=====+
  |
  | (Spark: clean, deduplicate)
  v
+=====+
| SILVER LAYER (Cleaned)      |
| - Validated, typed, deduped |
| - Delta Lake with versioning |
+=====+
  |
  | (Spark/dbt: aggregate, join)
  v
+=====+
| GOLD LAYER (Curated)       |
| - Business-ready aggregates |
| - Optimized for queries     |
+=====+
  |
  v
[Analytics / ML / Dashboards]

```

Project 11: End-to-End Streaming Pipeline

```

[Web Application (User Events)]
|
| (Event Collector)
v
[Kafka Topic: raw_clickstream]
|
+---> [Spark Job 1: Sessionization]
|      --> [Cassandra: session_store]
|
+---> [Spark Job 2: Real-time Aggregation]
|      --> [Druid: real_time_metrics]
|      --> [Grafana: Live Dashboard]
|
+---> [S3: raw event archive]
|      --> [Spark Batch: reconciliation]

```

Project 14: Data Pipeline with Full CI/CD

```

[Developer pushes code]
|
| (GitHub Actions triggered)
v
[CI Pipeline]
+-- Lint (sqlfluff, ruff)
+-- Unit Tests (pytest)
+-- Integration Tests
+-- dbt compile & test
+-- Great Expectations suite
|
| (All pass)
v
[CD Pipeline]
+-- Docker build & push
+-- Terraform apply
+-- Deploy Airflow DAGs
|
v
[Production Pipeline]
+-- Prometheus --> Grafana
+-- Airflow alerts --> Slack/PagerDuty

```

GENERIC DATA ENGINEERING PIPELINE (Reference Architecture)

[DATA SOURCES]	[INGESTION]	[STORAGE]
+++ Databases	-----> +++ Batch (Airflow)	--> +++ Data Lake (S3/GCS)
+++ APIs	-----> +++ Streaming (Kafka)	--> +++ Data Warehouse
+++ Files	-----> +++ CDC (Debezium)	--> +++ Database
+++ Streams	-----> +++ Webhooks	--> +++ Object Store
[PROCESSING]	[QUALITY]	[SERVING]
+++ Spark (batch)	+++ Schema validation	+++ BI Dashboards
+++ Spark Streaming	+++ Data tests (dbt)	+++ ML Feature Store
+++ dbt (SQL transforms)	+++ Great Expectations	+++ APIs
+++ Python scripts	+++ Anomaly detection	+++ Reports
[ORCHESTRATION]	[MONITORING]	[GOVERNANCE]
+++ Airflow / Prefect	+++ Prometheus/Grafana	+++ Data Catalog
+++ Scheduling	+++ Alerting (Slack/PD)	+++ Lineage
+++ Dependencies	+++ SLA tracking	+++ Access Control
+++ Retry logic	+++ Pipeline health	+++ Compliance